

Oracle PL/SQL MD5 & Package Learning Summary

Prepared by Ayub Miaze

1. Full Summary

Section	Description	Code / Notes
Tables	Tables used for storing users and logs.	<pre>CREATE TABLE APP_USERS (USER_ID NUMBER PRIMARY KEY, USERNAME VARCHAR2 (100), PASSWORD VARCHAR2 (100), STATUS VARCHAR2 (20)); CREATE TABLE APP_LOGS (LOG_ID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY, LOG_DATE DATE, ACTION_DESC VARCHAR2 (4000));</pre>
Package Spec	Declares procedures and functions for user management and logging.	<pre>CREATE OR REPLACE PACKAGE PKG_USER_MGMT AS PROCEDURE INS_USER (p_username IN VARCHAR2, p_password IN VARCHAR2, p_status IN VARCHAR2); PROCEDURE UPD_USER (p_user_id IN NUMBER, p_username IN VARCHAR2, p_status IN VARCHAR2, p_md5 IN VARCHAR2); FUNCTION GET_USER_MD5 (p_user_id IN NUMBER) RETURN VARCHAR2; PROCEDURE LOG_ACTION (p_desc IN VARCHAR2); END PKG_USER_MGMT;</pre>
Insert Procedure	Inserts a new user and logs the action.	<pre>PROCEDURE INS_USER (p_username IN VARCHAR2, p_password IN VARCHAR2, p_status IN VARCHAR2) IS BEGIN INSERT INTO APP_USERS (USER_ID, USERNAME, PASSWORD, STATUS) VALUES (APP_USERS_SEQ.NEXTVAL, p_username, p_password, p_status); LOG_ACTION ('Inserted user ' p_username); END;</pre>
MD5 Function	Returns MD5 checksum for concurrency control based on user data.	<pre>FUNCTION GET_USER_MD5 (p_user_id IN NUMBER) RETURN VARCHAR2 IS</pre>

Oracle PL/SQL MD5 & Package Learning Summary

Prepared by Ayub Miaze

		<pre>v_md5 VARCHAR2 (4000); BEGIN SELECT APEX_UTIL.get_hash (apex_t_varchar2 (USERNAME, PASSWORD, STATUS)) INTO v_md5 FROM APP_USERS WHERE USER_ID = p_user_id; RETURN v_md5; END;</pre>
Update Procedure	Checks MD5 checksum, locks the row (FOR UPDATE), updates if checksum matches, logs action.	<pre>PROCEDURE UPD_USER (p_user_id IN NUMBER, p_username IN VARCHAR2, p_status IN VARCHAR2, p_md5 IN VARCHAR2) IS v_md5_now VARCHAR2 (4000); BEGIN SELECT APEX_UTIL.get_hash (apex_t_varchar2 (USERNAME, PASSWORD, STATUS)) INTO v_md5_now FROM APP_USERS WHERE USER_ID = p_user_id FOR UPDATE; IF v_md5_now != p_md5 THEN RAISE_APPLICATION_ERROR (-20001, 'Data changed, please refresh.');</pre>
Autonomous Log Procedure	Logs actions independently using an autonomous transaction to avoid blocking main transaction.	<pre>PROCEDURE LOG_ACTION (p_desc IN VARCHAR2) IS PRAGMA AUTONOMOUS_TRANSACTION; BEGIN INSERT INTO APP_LOGS (LOG_DATE, ACTION_DESC) VALUES (SYSDATE, p_desc); COMMIT; END;</pre>
Example: Insert User Call	How to call the insert procedure.	<pre>BEGIN PKG_USER_MGMT.INS_USER('ayub', 'securepass', 'ACTIVE'); END;</pre>
Example: Get MD5 Checksum	Retrieve MD5 hash for a specific user.	<pre>DECLARE v_md5 VARCHAR2 (4000); BEGIN v_md5 := PKG_USER_MGMT.GET_USER_MD5(1); DBMS_OUTPUT.PUT_LINE(v_md5); END;</pre>
Example: Update with MD5 Check	Update user info only if data unchanged (MD5 matches).	<pre>DECLARE v_md5 VARCHAR2 (4000); BEGIN v_md5 := PKG_USER_MGMT.GET_USER_MD5(1); PKG_USER_MGMT.UPD_USER(1, 'ayub_miaze',</pre>

Oracle PL/SQL MD5 & Package Learning Summary

Prepared by Ayub Miaz

		'ACTIVE', v_md5); END;
Check Logs	View all logged actions.	SELECT * FROM APP_LOGS;

Some code snippets for better readability:

```
CREATE TABLE APP_USERS
(
    USER_ID      NUMBER PRIMARY KEY,
    USERNAME     VARCHAR2 (100),
    PASSWORD     VARCHAR2 (100),
    STATUS       VARCHAR2 (20)
);

CREATE TABLE APP_LOGS
(
    LOG_ID       NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    LOG_DATE     DATE,
    ACTION_DESC  VARCHAR2 (4000)
);

CREATE OR REPLACE PACKAGE PKG_USER_MGMT
AS
    PROCEDURE INS_USER (p_username  IN VARCHAR2,
                       p_password  IN VARCHAR2,
                       p_status    IN VARCHAR2);

    PROCEDURE UPD_USER (p_user_id   IN NUMBER,
                       p_username  IN VARCHAR2,
                       p_status    IN VARCHAR2,
                       p_md5       IN VARCHAR2);

    FUNCTION GET_USER_MD5 (p_user_id IN NUMBER)
        RETURN VARCHAR2;

    PROCEDURE LOG_ACTION (p_desc IN VARCHAR2);
END PKG_USER_MGMT;

PROCEDURE INS_USER (p_username  IN VARCHAR2,
                   p_password  IN VARCHAR2,
                   p_status    IN VARCHAR2)
IS
BEGIN
    INSERT INTO APP_USERS (USER_ID,
                          USERNAME,
                          PASSWORD,
                          STATUS)
        VALUES (APP_USERS_SEQ.NEXTVAL,
                p_username,
                p_password,
                p_status);

    LOG_ACTION ('Inserted user ' || p_username);
END;
```

Oracle PL/SQL MD5 & Package Learning Summary

Prepared by Ayub Miaz

```
FUNCTION GET_USER_MD5 (p_user_id IN NUMBER)
    RETURN VARCHAR2
IS
    v_md5    VARCHAR2 (4000);
BEGIN
    SELECT APEX_UTIL.get_hash (apex_t_varchar2 (USERNAME, PASSWORD, STATUS))
        INTO v_md5
        FROM APP_USERS
        WHERE USER_ID = p_user_id;

    RETURN v_md5;
END;

PROCEDURE UPD_USER (p_user_id    IN NUMBER,
                   p_username    IN VARCHAR2,
                   p_status      IN VARCHAR2,
                   p_md5         IN VARCHAR2)
IS
    v_md5_now    VARCHAR2 (4000);
BEGIN
    SELECT APEX_UTIL.get_hash (apex_t_varchar2 (USERNAME, PASSWORD, STATUS))
        INTO v_md5_now
        FROM APP_USERS
        WHERE USER_ID = p_user_id
    FOR UPDATE;

    IF v_md5_now != p_md5
    THEN
        RAISE_APPLICATION_ERROR (-20001, 'Data changed, please refresh.');

END IF;



UPDATE APP_USERS
        SET USERNAME = p_username, STATUS = p_status
        WHERE USER_ID = p_user_id;



LOG_ACTION ('Updated user ' || p_username);
END;



PROCEDURE LOG_ACTION (p_desc IN VARCHAR2)
IS
    PRAGMA AUTONOMOUS_TRANSACTION;
BEGIN
    INSERT INTO APP_LOGS (LOG_DATE, ACTION_DESC)
        VALUES (SYSDATE, p_desc);

    COMMIT;
END;



DECLARE
    v_md5    VARCHAR2 (4000);
BEGIN
    v_md5 := PKG_USER_MGMT.GET_USER_MD5 (1);


```

Oracle PL/SQL MD5 & Package Learning Summary

Prepared by Ayub Miaze

```
PKG_USER_MGMT.UPD_USER (1,  
                        'ayub_miaze',  
                        'ACTIVE',  
                        v_md5);  
  
END;
```

2. Key Concepts Recap

Concept	Explanation
MD5 Checksum	A hash value representing current data state, used to detect changes and prevent update conflicts.
FOR UPDATE	Locks the selected row to prevent other transactions from modifying it while current transaction works on it.
Autonomous Transaction	Allows logging to happen independently of main transaction, ensuring logs are saved even if main transaction fails.
Modular Procedures	Separate insert, update, get, and logging procedures for clear, maintainable code.

3. My Learnings

Learning Point	Details
Understanding MD5 Usage	MD5 hash is used to track row versions, preventing concurrent update conflicts by comparing before update.
Row Locking with FOR UPDATE	FOR UPDATE locks rows during a transaction to maintain data consistency and prevent race conditions.
Autonomous Transactions for Logging	Use autonomous transactions for logging to avoid blocking or rollback affecting main operations.
Modular Package Design	Organize code into modular packages separating insert, update, get, and log functionality for clarity.
Error Handling with Custom Exceptions	Use RAISE_APPLICATION_ERROR for clear, user-friendly error messages on concurrency conflict detection.
Using Apex Utility Functions	apex_util.get_hash and apex_t_varchar2 simplify creating MD5 hash from multiple columns easily.
Practical Use Cases for MD5	Ideal for optimistic concurrency control in multi-user environments to ensure safe data updates.
Commit inside Autonomous Transaction	Always commit inside autonomous transactions to make logs persistent even if main transaction fails.